

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20

Annexure A

Class Test

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
*I understand that any violation of academic integrity rules will result in disciplinary action. **Signature of the student:** K. Pujitha*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	

		strong analytical ability				
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Scenario: A drone is used to capture aerial images of a farmland to identify pest affected crop regions. Due to sensor limitations, wind movement, and changing sunlight, the captured images contain noise, shadows, and uneven illumination. Pest affected areas usually show visible texture and color variations compared to healthy crop regions.

Answer the following questions:

a) Preprocessing Pipeline — 4 Marks

Propose a suitable preprocessing pipeline to improve the quality of the farmland images before analysis. Your answer should include methods for noise reduction, illumination correction, contrast enhancement, and normalization. Justify the purpose of each step.

b) Feature Extraction — 3 Marks

Choose an appropriate feature extraction method to identify texture changes caused by pest damage. Explain why the selected method is suitable for detecting affected crop regions.

c) Rotation and Scale Variations — 3 Marks

Drone images may be captured from different heights and angles. Discuss how rotation and scale variations can affect feature detection. Suggest suitable methods to make the feature extraction process more robust.

Question 2: Stereo Vision and Motion Estimation

Scenario: A self-driving car is equipped with stereo cameras to navigate through a busy street. The system must estimate the distance of nearby vehicles and pedestrians in real time. It should also track moving objects accurately to avoid collisions, especially when pedestrians or vehicles change speed suddenly.

Answer the following questions:

a) Stereo Disparity and Depth Estimation — 4 Marks

Explain how stereo cameras use the difference between left and right image views to estimate the depth of objects. Discuss how disparity changes for nearby and distant objects.

b) Optical Flow for Motion Tracking — 3 Marks

If a pedestrian suddenly changes speed or direction, explain how optical flow helps in tracking the pedestrian's motion between consecutive video frames.

c) Errors and Mitigation Techniques — 3 Marks

Identify possible sources of error in depth estimation and motion tracking, such as camera misalignment, poor lighting, occlusion, fast motion, or textureless regions. Suggest suitable techniques to reduce these errors.

1) Image Preprocessing and Feature Extraction

Given Scenario

- A drone captures aerial images of farmland to identify pest-affected crop regions.
- The captured images contain:
 - Noise due to sensor limitations.
 - Shadows due to changing sunlight.
 - Uneven illumination.
 - Texture and color variations between healthy and affected crops.
- The objective is to preprocess the images and extract useful features for pest detection.

(a) Preprocessing Pipeline

Answer

Drone images captured under outdoor conditions may contain noise, lighting variations, and contrast problems. A suitable preprocessing pipeline improves image quality before feature extraction.

The preprocessing pipeline is:

Input Drone Image → Noise Reduction → Illumination Correction → Contrast Enhancement → Image Normalization → Feature Extraction

1. Noise Reduction

Method Used: Gaussian Filter / Median Filter

Sensor noise caused by the drone camera can affect the quality of crop images.

- **Gaussian filter** is suitable for removing random noise.
- **Median filter** can remove impulse noise while preserving edges.

Purpose

- Removes unwanted noise.
- Prevents false detection of pest regions.
- Improves image clarity.

2. Illumination Correction

Method Used: Homomorphic Filtering or Background Correction

Since sunlight changes create shadows and uneven brightness, illumination correction is required.

Purpose

- Removes effects of shadows.
- Makes brightness uniform throughout the image.
- Helps distinguish pest-affected regions from healthy crops.

3. Contrast Enhancement

Method Used: CLAHE (Contrast Limited Adaptive Histogram Equalization)

Pest-affected areas may have small intensity differences compared to healthy plants.

Purpose

- Enhances local contrast.
- Highlights texture and color differences.
- Improves visibility of damaged crop regions.

4. Image Normalization

Method Used: Intensity Normalization

Pixel values are converted into a standard range such as 0–255.

Purpose

- Reduces variation between images.
- Provides consistent input for feature extraction.
- Improves accuracy of classification algorithms.

Conclusion

The above preprocessing steps improve image quality by reducing noise, correcting lighting variations, enhancing crop details, and preparing the image for accurate pest detection.

(b) Feature Extraction

Answer

A suitable feature extraction method for detecting pest-affected crop regions is Local Binary Pattern (LBP).

Local Binary Pattern (LBP)

LBP is a texture-based feature extraction technique used to analyze local patterns in an image.

Working Principle

- Each pixel is compared with its neighboring pixels.

- The comparison generates a binary pattern.
- These patterns represent texture information.

Why LBP is Suitable

Pest damage changes the texture of crop surfaces, such as:

- Leaf spots.
- Irregular patterns.
- Surface damage.
- Changes in leaf structure.

LBP can identify these texture variations effectively.

Advantages

- Simple and computationally efficient.
- Robust to small illumination changes.
- Suitable for texture analysis.

Therefore, LBP is suitable for identifying pest-affected crop regions based on texture differences.

(c) Rotation and Scale Variations

Answer

Drone images may be captured from different heights and angles, causing changes in object size and orientation.

Effect of Rotation Variation

Rotation changes the orientation of crop patterns.

Problems

- Extracted features may appear different.
- Matching accuracy decreases.
- Same crop region may be incorrectly classified.

Effect of Scale Variation

Scale changes occur when the drone flies at different heights.

Problems

- Crop regions appear larger or smaller.
- Feature size changes.
- Important details may be lost.

Methods to Improve Robustness

1. SIFT (Scale-Invariant Feature Transform)

- Provides scale and rotation invariance.
- Detects important keypoints at different sizes and orientations.

2. SURF (Speeded-Up Robust Features)

- Faster alternative to SIFT.
- Handles scale and rotation changes effectively.

3. Data Augmentation

Training images can be modified by:

- Rotation.
- Scaling.
- Flipping.

This helps machine learning models learn different variations.

2) Stereo Vision and Motion Estimation

Given Scenario

- A self-driving car is equipped with stereo cameras.

- The system must:
 - Estimate the distance of nearby vehicles and pedestrians.
 - Track moving objects in real time.
 - Avoid collisions even when pedestrians or vehicles change speed suddenly.
- The objective is to estimate depth accurately and perform reliable motion tracking.

(a) Stereo Disparity and Depth Estimation

Answer

A stereo vision system uses two cameras placed a small distance apart, similar to human eyes. Both cameras capture the same scene from slightly different viewpoints.

The position of an object appears different in the left and right images. This difference is called disparity.

Working

1. The left and right cameras capture images simultaneously.
2. The system identifies the same object in both images.
3. The horizontal shift between the two images is calculated.
4. Using this disparity, the distance (depth) of the object is estimated.

The depth of an object is inversely proportional to the disparity and is calculated using:

$$\text{Depth} = \frac{f \times B}{d} \quad \text{Depth} = \frac{f \times B}{d}$$

where,

- **f** = Focal length of the camera
- **B** = Distance between the two cameras (baseline)
- **d** = Disparity

Disparity for Nearby and Distant Objects

- Nearby objects have large disparity because their positions differ more between the two images.

- Distant objects have small disparity because they appear almost at the same position in both images.

Thus, stereo vision enables a self-driving car to estimate the distance of vehicles and pedestrians accurately for safe navigation.

(b) Optical Flow for Motion Tracking

Answer

Optical flow is used to estimate the movement of objects between two consecutive video frames.

When a pedestrian suddenly changes speed or direction, the optical flow algorithm tracks the movement of feature points such as edges or corners.

Working

1. Detect important feature points on the pedestrian.
2. Compare their positions in consecutive frames.
3. Calculate the displacement of each feature point.
4. Generate motion vectors representing the pedestrian's movement.

Effect of Sudden Speed or Direction Change

- If the pedestrian moves faster, the motion vectors become longer.
- If the pedestrian slows down, the vectors become shorter.
- If the pedestrian changes direction, the orientation of the vectors also changes.

By continuously updating these motion vectors, optical flow helps the self-driving car track pedestrians accurately and predict their movement to avoid collisions.

(c) Errors and Mitigation Techniques

Answer

Several factors can affect the accuracy of depth estimation and motion tracking.

1. Camera Misalignment

If the stereo cameras are not properly aligned, incorrect disparity values are obtained, leading to inaccurate depth estimation.

Solution:

Perform camera calibration to correctly align the stereo cameras.

2. Poor Lighting

Low light or strong shadows reduce image quality and make feature detection difficult.

Solution:

Apply image preprocessing techniques such as illumination correction and contrast enhancement.

3. Occlusion

An object may be partially hidden behind another object, making feature matching difficult.

Solution:

Use Kalman Filter or object tracking algorithms to predict the object's position until it becomes visible again.

4. Fast Motion

Rapid movement may produce blurred images and incorrect optical flow vectors.

Solution:

Use high-frame-rate cameras and faster tracking algorithms.

5. Textureless Regions

Plain surfaces without sufficient texture make it difficult to find matching points.

Solution:

Use robust feature extraction methods such as SIFT, SURF, or ORB to detect reliable keypoints.

